problems/MHD/brioWu1d

DynSGS and positivity setup, as coded on branch `sm/MHD` (commit `dcc3653`)

How to read the underlines. Solid underline: the setting (or the code path) is *identical in every case of `problems/MHD` to which it applies* — all nine cases for general settings, the eight `DSGS_MHD` cases for DynSGS settings, the three `:positivity_method => "conservative"` cases for the element limiter. Dashed underline: shared by a *majority* of those cases but not all of them. No underline: specific to this case or to a minority. The cross-case comparison is in `summary.pdf` of this directory.

Scope. Everything below is read off the deck (`user_inputs.jl`), the case files and the kernels they reach on `sm/MHD`; defaults are the ones `mod_inputs.jl` fills in when the deck is silent, and environment overrides are taken at their unset values (`JEXPRESSO_BW_*`). Comments in the files were not taken as evidence where the code says otherwise; the disagreements found are listed in Section 6.

1 The problem as set up

Deck key / item	Value as run	Source / remark
Equations	1D ideal MHD, state $(\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z)$, 8 slots, no GLM field	`initialize.jl`, `user_flux.jl`
EOS	$p = (\gamma-1)(E - \frac{1}{2}\lvert\mathbf{m}\rvert^2/\rho - \frac{1}{2}\lvert\mathbf{B}\rvert^2)$, $\gamma = 2$	`user_flux.jl` (γ_{mhd} `= 2.0`)
Domain, mesh	$x \in (0,1)$, 150 elements (600 DOFs)	`:nelx` $= 600/N$ (`JEXPRESSO_BW_DOFS`, `_NELX`)
Discretization	CG spectral elements, LGL nodes, lumped (diagonal) mass	`:interpolation_nodes => "lgl"`
Polynomial order	$N = 4$	`:nop` (`JEXPRESSO_BW_NOP`)
Time integration	`CarpenterKennedy2N54`, fixed Δt, $\Delta t = 5 \times 10^{-5}$, $t_{end} = 0.1$	$\Delta t = 5 \times 10^{-5} \cdot 600/(n_{el}N)$
Variables	`TOTAL()`, `:energy_equation => "energy"`	
Initial data	left $(\rho, u, p, B_y) = (1, 0, 1, 1)$, right $(0.125, 0, 0.1, -1)$, $B_x = 0.75$, jump at $x = 0.5$	`initialize.jl`
Boundary conditions	Dirichlet on all 8 slots at both ends (the left/right states)	`user_bc.jl`
Sources	none (`:lsource => false`)	`user_source.jl`
Other switches	`:lfilter`, `:lkep`, `:entropy_variables`, `:lrichardson`, `:ladapt` all false	deck / defaults

2 DynSGS: deck keys and effective values

The deck selects `:visc_model => DSGS_MHD()` with `:lvisc => true`, which in 1D reaches `compute_dsgs_viscosity!(::DSGS_MHD, ::NSD_1D)` (`SGS.jl:1929`) through `viscous_rhs_el!(..., ::NSD_1D)` (`rhs.jl:1398`).

The deck rescales the three constants by $h_{fac} = (N+1)/N = 5/4$ (`JEXPRESSO_BW_HSCALE` unset ⇒ `"nop"`): the kernel uses $\Delta = \Delta x/(N+1)$, and the scaling turns it into Dao & Nazarov's $\Delta x/N$.

Deck key / item	Value as run	Source / remark
`:visc_model`	`DSGS_MHD()`	
`:`μ (slot multipliers)	all 1 (8 entries)	mass diffusion is on
`:dsgs_sensor`	`"residual"`	element residual, stage-consistent stencil
`:dsgs_CR`	$1.0 \cdot h_{fac}^2 = 1.5625$	$\equiv C_R = 1$ with $\Delta x/N$
`:dsgs_Cmax`	$0.5 \cdot h_{fac} = 0.625$	$\equiv C_{\max} = 0.5$ with $\Delta x/N$
`:dsgs_Cmin`	$0.06 \cdot h_{fac} = 0.075$	$\equiv C_{\min} = 0.06$ with $\Delta x/N$ (`JEXPRESSO_BW_CMIN`)
`:dsgs_gamma`	$= \gamma_{mhd} = 2$	
`:dsgs_Prt`	1.0	no effect in the conserved form
`:dsgs_rel`	1.0 (default)	floor = the physical scale itself
`:dsgs_norms`	`"domain"`	means/spreads over the whole tube (MPI-reduced)
`:dsgs_conserved`	true	ν on every slot
`:dsgs_nazarov_energy`	false (default)	
`:dsgs_fast_floors`	false (default)	
`:dsgs_thermal_residual`	false (default)	2D only in any case
`:dsgs_hold_steps`	2	`JEXPRESSO_BW_HOLD`
`:dsgs_cutoff`	0	`JEXPRESSO_BW_CUTOFF`
`:dsgs_freeze_stage`	false at `dcc3653`; true since `b956ccd` (default)	at `dcc3653` recomputed at every RK stage; now once per step
`:ldsgs_nodal`	false (element form)	`JEXPRESSO_BW_NODAL`; `:dsgs_Cl => 0.4` only acts in the nodal form
`:dsgs_nodal_rho`, `:dsgs_ref_weight`, `:dsgs_reference`	false	defaults
`:dsgs_local_rel`	1.0 (default)	unused (domain norms)

3 DynSGS as it runs for this case

Notation: element K, LGL node i, element lumped mass $m_i^K = \omega_i J_{K,i}$, $\Delta_K = \Delta x/(N+1) = 1/750$, stage time $t = t^n + \tau$, $h = \Delta t$.

1. **History.** At the first RHS call of every step ($t - t_{hist} \geq 0.999\,\Delta t$) the DynSGS triple is rotated, $(\mathbf{q}^{n-2}, \mathbf{q}^{n-1}, \mathbf{q}^n) \leftarrow (\mathbf{q}^{n-1}, \mathbf{q}^n, \mathbf{q})$, and the rotation counter n_{hist} is incremented (`rhs.jl:936`). The triple is seeded with the initial condition.

2. **Hold.** With $H = 2$ the residual right-hand side and the weights are held at zero while $n_{hist} < \max(2, H) + 1 = 3$, i.e. during the first two steps; ν is then the $C_{\min}$ floor alone (`rhs.jl:182, 195, 947`).

3. **Time derivative** (residual sensor, `rhs.jl:964-971`):

$$D_t q = \begin{cases} \dfrac{3q^n - 4q^{n-1} + q^{n-2}}{2h}, & \tau \leq 10^{-8} h, \\ \dfrac{2\tau + h}{\tau(\tau + h)}\, q(\tau) - \dfrac{\tau + h}{\tau h}\, q^n + \dfrac{\tau}{h(\tau + h)}\, q^{n-1}, & \tau > 0. \end{cases}$$

4. **Residual** of slot s at node i of K: $R^s_{K,i} = \left| D_t q_i^s - r_i^{K,s}/m_i^K \right|$, with r^K the element's own weak inviscid right-hand side (`params.rhs_el`, not assembled). **Dirichlet nodes:** the two end nodes are listed by `_dsgs_boundary_pairs!` and their entries replaced by $m_i^K D_t q_i$, so $R \equiv 0$ there for all 8 slots (`rhs.jl:313, 346`).

5. **Normalization** (domain): $\langle q^s \rangle$ is the plain average over all element nodes of the tube (shared nodes counted once per element, MPI-summed), $S^s = \max |q^s - \langle q^s \rangle|$ (MPI-max). With $\bar\rho = \max(|\langle \rho \rangle|, 10^{-16})$, $\bar p = (\gamma - 1)\max(\langle E \rangle - |\langle \mathbf{m} \rangle|^2/(2\bar\rho) - \tfrac{1}{2}|\langle \mathbf{B} \rangle|^2, 0)$ and the sound speed $\bar c = \sqrt{\gamma \bar p/\bar\rho}$ (no fast floors),

$$n^s = \max\left(S^s,\ r\,\sigma^s\right) + 10^{-16}, \qquad \sigma^\rho = \bar\rho,\ \ \sigma^{\rho u, \rho v, \rho w} = \bar\rho \bar c,\ \ \sigma^E = \bar\rho \bar c^2,\ \ \sigma^B = \sqrt{\bar\rho}\,\bar c,$$

with $r =$ `:dsgs_rel` $= 1$ (in the code: $10^3 \cdot$ `:dsgs_rel` times the built-in 10^{-3}).

6. **Element ratio and cap speed.** $\mathcal{R}_K = \max_{i\in K}\max_{s\le 8} R^s_{K,i}/n^s$ (all 8 slots; there is no ψ), $w_K = \max_{i\in K}\big(|\mathbf{v}_i| + c_{f,i}\big)$ with the *exact fast speed along* x, $c_f^2 = \frac{1}{2}\big(a^2+b^2+\sqrt{(a^2+b^2)^2 - 4a^2b_x^2}\big)$, $a^2 = \gamma p/\rho$, $b^2 = |\mathbf{B}|^2/\rho$, $b_x^2 = B_x^2/\rho$, p clipped at 0 (`SGS.jl:2075-2088`).
7. **Coefficient** (`SGS.jl:2092-2096`):

$$\nu_c = \max\big(0, \min(C_{\max}\Delta_K w_K,\ C_R\Delta_K^2\mathcal{R}_K)\big), \qquad \nu_{\text{fl}} = C_{\min}\Delta_K w_K, \qquad \nu_K = \max(\nu_c, \nu_{\text{fl}}),$$

one value per element, recomputed at every Runge–Kutta stage; $(C_R, C_{\max}, C_{\min}) = (1.5625, 0.625, 0.075)$ with $\Delta_K = \Delta x/5$, i.e. (1, 0.5, 0.06) on $\Delta x/4$.
8. **Slot coefficients** (conserved form): $\mu_K^s = \mu[s]\,\nu_K = \nu_K$ for all 8 slots.

4 The viscous operator

`user_primitives.jl` returns the conserved variables themselves, and the 1D loop applies one scalar flux per slot (`rhs.jl:2823`, `1349`):

$$\partial_t\mathbf{q} + \partial_x\mathbf{F}(\mathbf{q}) = \partial_x\big(\nu_K\,\partial_x\mathbf{q}\big) \quad \text{on } (\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z),$$

a Laplacian of the conserved state, in divergence form, with no separate $\tau\cdot\mathbf{v}$ or resistive-work term (the E Laplacian carries them). ν is piecewise constant (a staircase with jumps at element interfaces). B_x is constant in the exact solution; its residual stays in the max and acts as an oscillation detector.

5 Positivity

- **No state limiter.** `:lpositivity` is false (default) and the solver carries no stage limiter: nothing modifies the state to enforce $\rho > 0$ or $p > 0$.
- **Flux-only floor.** `pressure_mhd1d` returns $\max\big((\gamma-1)(E-\dots), 10^{-9}\big)$ (`const p_floor_mhd = 1.0e-9`, `user_flux.jl`). It is used by `user_flux!` and by the output; it changes the flux of a node whose pressure has gone below 10^{-9}, not the state.
- **What keeps the solution admissible** is therefore the dissipation alone: the conserved-form Laplacian with $\nu \ge \nu_{\text{fl}} = 0.075\,\Delta_K w_K > 0$ everywhere (a background of $C_{\min}/C_{\max} = 12\%$ of the cap), and the cap $C_{\max}\Delta w$ at the discontinuities. Inside DynSGS itself the pressure is clipped at 0 when it builds $\bar{p}$ and c_f, so a negative nodal p cannot produce a NaN coefficient.

6 Observations: code versus comments

1. [**Fixed in `23e4974`.**] **At `dcc3653` the default (element-form) run failed on its first DynSGS call.** Commit `708bd78` added the keyword `lthermal = ...` to the 1D call (`rhs.jl:1456`) but not to the signature of `compute_dsgs_viscosity!(::DSGS_MHD, ::NSD_1D)` (`SGS.jl:1929-1956`, last keyword `lfast_floors`). Julia 1.11.5, calling the kernel's own source with the exact keyword set of `rhs.jl:1434-1456`, raised `MethodError: ... got unsupported keyword argument "lthermal"`. The nodal path (`JEXPRESSO_BW_NODAL=1`) does not pass `lthermal` and was not affected. `23e4974` adds `lthermal::Bool=false` to the 1D signature; the thermal residual is 2D-only, so the keyword is ignored there, and the same call returns a bit-identical coefficient with `lthermal` omitted, false or true.
2. The header of `user_inputs.jl` says "integrated with SSPRK53"; the deck sets `CarpenterKennedy2N54()`.
3. `DSGS.md` §4.7 says this case needs a 3% $C_{\min}$ floor; the deck sets $0.06\,h_{fac}$ (its own comment says 6%; relative to the cap $C_{\max}$ that is 12%).
4. [**Changed in `b956ccd`.**] `:dsgs_freeze_stage` now defaults to true for `DSGS_MHD` with the residual sensor: the coefficient is computed once per step, at the stage on t^n, and held over the stages, which removes the $O(\Delta t)$ floor the RK internal stages put under the residual. Checked on this case, frozen against per stage ($N = 4, 7$; 600 and 1200 DOFs): with the deck's floor the density errors differ by under 1% over the tube and under 5% in the fan; without it (`JEXPRESSO_BW_CMIN=0`) by under 3%, with slightly less excess total variation frozen; ρ bounds unchanged. `JEXPRESSO_BW_FREEZE=0` gives the per-stage run.

Appendix: code map (branch `sm/MHD`, commit `dcc3653`)

Mechanism	Where (file:line)
Deck defaults of every `:dsgs_*` key	`src/io/mod_inputs.jl:1085-1376`
Deck defaults of `:lpositivity`, floors, report `:positivity_method`; rebuild of the RK solver	`src/io/mod_inputs.jl:456-477` `src/io/mod_inputs.jl:828-838`
DynSGS history rotation, stage weights, hold, freeze	`src/kernel/operators/rhs.jl:927-984`
Stencil selection `_dsgs_stencil`	`rhs.jl:141`
Hold length `_dsgs_hold_steps` ($\max(2, H) + 1$ rotations)	`rhs.jl:182`
Residual right-hand side (element / legacy / Dirichlet)	`rhs.jl:188-280`; legacy fill `rhs.jl:282`; boundary zeroing `rhs.jl:313, 346`
1D `DSGS_MHD` dispatch	`rhs.jl:1398-1470`
2D `DSGS_MHD` dispatch (sets `dsgs_ref_weight`, `dsgs_split_energy`, `dsgs_nodal_rho`)	`rhs.jl:1517-1610`
2D viscous assembly `_viscous_rhs_el_2d_dsgs!`	`rhs.jl:1796`
2D weak viscous flux `_expansion_visc!` (conserved flag, ref. weight, split energy, $\tau \cdot \mathbf{v}$, resistive work)	`rhs.jl:2943-3210` (`lcons` 2959, weight 2978, split 3050, $\tau \cdot \mathbf{v}$ 3076, $\eta \mathbf{B} \cdot \nabla \mathbf{B}$ 3091)
1D weak viscous flux	`rhs.jl:2823`; element loop `rhs.jl:1321`
2D `DSGS_MHD` element kernel	`src/kernel/physics/SGS.jl:1598-1912`
1D `DSGS_MHD` element kernel	`SGS.jl:1929-2120`
Nodal kernels (off in every MHD deck; BW can switch it on with `JEXPRESSO_BW_NODAL=1`)	`SGS.jl:2144` (1D), `SGS.jl:2515` (2D)
Normalization floor `_dsgs_denom`, `:dsgs_rel` multiplier	`SGS.jl:676-679`; set per call in `rhs.jl:192`
Smoothness cutoff `_dsgs_cut`	`SGS.jl:703`
`SGS_diffusion(::DSGS_MHD)` (nodal-ρ switch)	`SGS.jl:547-569`
Element length Δ_K^{elem} (2D: min of corner extents)	`src/kernel/mesh/mesh.jl:6430-6457`; 1D Δx: `mesh.jl:5863`
`uprimitive` width ($n_{eqs} + 2$ in 2D, $n_{eqs} + 1$ in 1D)	`src/kernel/globalStructs.jl:139-153`
Realizability repair (`"repair"`, unused by the MHD decks)	`src/kernel/positivity/Positivity.jl:294`; driver `positivity_driver.jl:124`; called in `rhs.jl:821`
Conservative element limiter (`"conservative"`)	`Positivity.jl:433-559`; driver `positivity_driver.jl:245-337`
Main `solve` call (`saveat` non-empty ⇒ `calck = true`)	`src/kernel/solvers/TimeIntegrators.jl:855`

Stage-limiter calls in `CarpenterKennedy2N54`: with `williamson_condition = false` the limiter is called before the right-hand side of stages 2–5 and once after stage 5 (OrdinaryDiffEq v6.90/v6.95, `low_storage_rk_perform_step.jl`). In OrdinaryDiffEq `calck = true` forces `williamson_condition = false` in the 2N cache, and Jexpresso's `solve` passes a non-empty `saveat`, so this holds for every deck that hands the solver a stage limiter. (Jexpresso ships no `Manifest.toml`; the package version is whatever the environment resolves, and the check above was made against v6.90/v6.95.)